



# Daniil Parshin

C++ Software Engineer | Systems & Performance

## Professional Summary

C++ Software Engineer focused on systems and performance engineering on Linux. I develop and optimize performance-sensitive C/C++ and Python software for profiling, graphics, and CPU inference, with hands-on experience in multithreading, benchmarking, build systems, CI/CD, debugging, and automated testing.

## Experience

### Huawei — Nizhny Novgorod, Russia

#### Software Engineer — Profiling Tools Team | November 2025 — Present

- Refactored the CMake superbuild of a performance profiling product comprising dozens of components, improving build speed by ~30% and decoupling package assembly from component builds.
- Maintain C/C++ and Python product code and Jenkins pipelines; automate builds, artifact handling, packaging, and Docker-based build/test environments.
- Develop unit tests with GoogleTest and integration tests with Robot Framework; automate test execution and reporting in CI.

#### Software Engineer — GPU Team | June 2024 — November 2025

- Improved BC1 texture compression quality by ~3% through dithering techniques and integrated ISPC-accelerated BC7 compression into the Mesa graphics driver; verified both through quality and performance benchmarks.
- Profiled OpenGL workloads on Linux using gdb and trace analysis; built Python/Bash tooling for automated quality evaluation and benchmarking.
- As a parallel cross-team project, developed IDet, a C++17 CPU inference library for text, face, and clothing detection using OpenCV, ONNX Runtime, and OpenMP.
- Optimized tiled inference and threading to **p99 below 10 ms per detector on FHD frames** on Kunpeng 920B.
- Built a Python pipeline for dataset preparation, YOLO training, evaluation, and ONNX export over **270k** images.

## Additional Experience

### Resident and Ambassador — NEIMARK IT-Academy | September 2022 — September 2024

Delivered team software projects, mentored hackathon participants, and contributed to a winning AI ethics project. [Program Page](#) | [Certificates](#)

### Private Tutor — Mathematics and Computer Science | September 2022 — Present

Prepared 35+ students for mathematics and computer science exams; taught algorithms and adapted learning plans to individual needs.

## Projects

### IDet — CPU ROI detection library | [GitHub](#)

CPU-only library for ROI detection. Stack: C++17, ONNX Runtime, OpenCV, OpenMP, Meson.

### YOLO Training and Deployment Pipeline | [GitHub](#)

End-to-end pipeline for dataset preparation, training, evaluation and ONNX deployment. Stack: Python, Ultralytics YOLO, ONNX Runtime, Bash.

### iVoice — Local Audio and Text Command Processing | [GitHub](#)

Local-first pipeline for speech recognition and command normalization. Stack: Python, faster-whisper, Transformers, FastAPI.

## Awards and Training

### Alice & Yandex Smart Devices — Yandex × HSE × MIPT | April 2026

Completed a 72-hour intensive on smart-device architecture, embedded systems, and resource-constrained ML deployment; developed and defended a team R&D project. [Certificate](#)

### Hack Lab — Sber × School 21 | November 2025

Won 1st place with the team in an HPC hackathon covering parallelization, OpenMP, hardware-level optimization, random number generation, and data integrity. [Certificate](#)

## Details

Nizhny Novgorod, Russia

## Contacts

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## Links

[GitHub](#) | [LeetCode](#) | [Telegram](#)

## Technical Skills

### Programming

C++, C, Python, Bash; algorithms & data structures

### Build, CI/CD & Testing

CMake, Meson, Jenkins, Docker, Git, GTest, Robot Framework

### Tools & Collaboration

Jira, Confluence

### Systems & Performance

Linux, gdb, NUMA, profiling, benchmarking, trace analysis, OpenMP, multithreading

### Graphics

Mesa, OpenGL, BCn

### Computer Vision & ML

OpenCV, ONNX Runtime, YOLO, DBNet, SCRFD, model training & export, CPU / GPU inference

## Languages

Russian — Native

English — B1

Spanish — A1

**YADRO Winter School — Programming for RISC-V** | February 2025

Completed intensive training in RISC-V programming, computer architecture, and low-level systems with practical project work. [Certificate](#)

**Samsung IT-School — Mobile Development** | September 2019 — June 2020

Studied Android, Java, OOP, algorithms, and data structures. Built a Firebase messenger and won 1st place in the regional round. [Certificate](#)

 **Education**

**HSE University — Nizhny Novgorod, Russia** | September 2026 — Present

Master's Degree in Applied Mathematics and Information Science. Program: Data Mining

**HSE University — Nizhny Novgorod, Russia** | September 2022 — June 2026

Bachelor's Degree in Applied Mathematics and Information Science | [Thesis](#)